Joint Quantities, Vehicle Ownership, and Physical Realization

A bounded resource study accompanying stable GI-TBC

1 From inventory prices to physical fleets

The stable Gini-interval tailored branch-and-cut method (GI-TBC/B, S2/ms2) retains complete improving-domain coverage, valid cutoffs and a scope-aware proof ledger. E+ retains RN16384 and the inherited proof-only consumer. The present study is an isolated primal resource experiment. Neither a resource witness nor an inner-domain bound is silently consumed by the complete exact engine.

With the original binary64 input coefficients, write $r_i = c_i Y_i$, $S = \sum_i r_i$, and, for $S > 0$,

$$G(Y) = \frac{\sum_{i<j} |r_i - r_j|}{nS}, \quad H(Y) = \sum_i \omega_i |r_i - 1|, \quad J(Y) = G(Y) + H(Y).$$

The coefficient ω_i includes λ. The repaired independent zero branch is retained; the two studied RN scopes have a qualified positive stock denominator and $\gamma_L = 0$. All comparisons bind the same inventory supports, original improvement scope, weights and numerical contract.

Let D_I denote the inherited integer inventory relaxation. For RX35, D_F removes precisely the original 26 vectors excluded by the R157 necessary-resource certificates; for NB35, $D_F = D_I$. Let $D_R(T)$ project unique legal station service, eligible integer ownership, per-car balances, return capacity, complete handling and necessary tours of at most three stations. Let $D_P(T)$ be the physical fleet projection. In the inherited qualified contract,

$$D_P(T) \subseteq D_R(T) \subseteq D_F \subseteq D_I, \qquad F_I \le F_F \le F_R \le F_P.$$

The middle inclusion uses the R157 resource exclusions and R158's actual C++ duration/budget qualification, rather than physical impossibility alone. Later finite exclusions do not redefine D_F.

An independently verified integer resource point gives U_R and hence

$$0 \le F_R - F_I \le U_R - L_I, \qquad \eta_R = \frac{U_R - L_I}{U_P^{\mathrm{ref}} - L_I}.$$

The reference U_P^{ref} is held fixed for each role. This scalar interval indicator is not realized complete-algorithm gap reduction. If $D_R \subseteq D_C \subseteq D_I$, then $F_C \le F_R \le U_R$. This statement does not cap the joint benefit of cuts in a full routing relaxation Q: the resource point may have no compatible full lift into Q. Likewise, $U_P - U_R$ is a witness-price difference, not a lower bound on $F_P - F_R$.

2 A bounded simultaneous inner domain

For vehicle k, let P_k and D_k be cumulative pickups and deliveries. Empty departure gives return $R_k = P_k - D_k$, with $0 \le R_k \le Q_k$. Including return unloading,

$$t_p P_k + t_d (D_k + R_k) = (t_p + t_d) P_k.$$

Cumulative P_k may exceed Q_k. Only each single station operation and the return are bounded by vehicle capacity in the resource model; physical prefix loads are checked separately.

The short fixed-owner reference optimizes original J over full supported quantities on the initial disjoint owner sets A_k. It allows zero service and direction changes, while initially unassigned stations retain $Y_i = b_i$. A full-A_k three-stop budget remains conservative after zeroing a station. This is an inner restriction, not exhaustive fixed-owner quantity search. The original RN centering/scale and one-sided envelope are retained.

The joint mechanism releases all supported quantities and eligible owners on eight stations at once. Every outside quantity and nonzero owner remains fixed, with its full pickup, delivery and return contribution included. Conditional active-subset budgets are rebuilt for every vehicle. Activation, zero service, direction changes, cross-car ownership and unequal-net quantity/ownership compensation are permitted without inheriting an uncovered old active-set budget.

1. Score station i by its best unconstrained single-stock reduction in true J, multiplied by one plus current owner handling pressure.

2. Choose the step-ranked anchor, then seven stations by that score plus anchor-weighted geometric proximity; break ties by station index.

3. Solve the joint original-J subproblem with a complete feasible Start, Seed 0, Threads 1 and at most 512 nodes. Retain strict true-J improvements and read them back independently in the complete original resource domain.

4. Stop after 12 blocks or the ordinary whole-method deadline. No nondecreasing intermediate move or time-triggered algorithm switch is used.

Both roles use the same rule and capacities. The short references use 2048 nodes each. The declared domain can coordinate eight changes beyond one single-stock change, owner move/swap or pair transfer. This is not a reachability theorem against every sequence of historical moves. Trace evidence distinguishes allowed freedoms from those actually exercised. Every subproblem lies inside D_R. Its feasible point can give a global resource upper; its bound, optimum or failure remains local. None enters the global proof ledger.

The new repeated model construction, block selection, callback and completion are C++. The standalone tool loads an explicitly exported, unchanged qualified RN objective; this offline boundary is not represented as a fully integrated production candidate. With $s = \sum_i |\mathcal{Y}_i|$, the new model has $O(Ms + n^2)$ columns and $O(Mn^3 + s + n^2)$ rows. A node limit does not bound presolve time or establish polynomial complexity.

3 Resource improvements and physical obstructions

Role	U_P^{ref}	Fixed-owner U_R	Joint U_R	η_R
RX35	.2996694007	.2945369423	.2911853932	.86663993
NB35	.4502915033	.4258556867	.4235630180	.50808063

The RX35 fixed-owner start is the inherited review-side resource witness. Its short reference uses 59.078 seconds without a new accepted point. NB35's reference uses 17.581 seconds and accepts seven quantity improvements. The 12-block joint calls use 4.765/5.879 outer seconds and 31/246 total nodes, with two/four accepted improvements, respectively. Every block returns normally with

Native OPTIMAL; this numerical status is not an unguarded exact original-J optimum. Original fractions and complete raw-resource lifts qualify the reported uppers independently.

RX35 improves by .0033515491 beyond its short fixed-owner reference; NB35 improves by .0022926687. Both trajectories contain nonzero quantity-net changes on old owners compensated by ownership changes. NB35 also activates station 26. No accepted zero-service or direction-flip event is observed, although both are open in the declared domain.

One post-trace same-block RX35 contrast fixes owner without adding a full-A budget. It finishes at 21 nodes in 1.020 outer seconds. Its guarded, transported restricted original-J lower is .2945344157494531, while the joint event attains .2922946074975457 by changing quantities at stations 19/35 and moving ownership of station 25. Owner freedom is necessary for that improvement in that block and outside state. This does not close all fixed-owner quantities or prove impossibility under every historical move sequence.

Role	Car	P_k	Unloaded tour	Prefix extra	Total time
RX35	0	13	5335.219	0.000	6895.219
	1	16	3986.951	241.486	6148.436
	2	25	3412.088	0.000	6412.088
NB35	0	32	3470.606	0.679	7311.285
	1	29	4052.116	482.803	8014.919
	2	14	5740.652	74.377	7495.029

These are complete fixed-Y/owner DP minima at the joint best points; $T = 6300/7200$ for RX35/NB35. Every car has $P_k = D_k$ and zero return. The RX35 car 2 obstruction matches the inherited review-side explanation. The new car 0 active set yields a pure-tour pickup budget of 8 instead of the 13 used by the resource point. RX35 car 1 has a genuine prefix penalty but still fits. NB35 car 1 is over time even without prefixes; prefixes add 482.803 seconds and reduce its fixed-combination pickup budget from 26 to 22. All three NB35 joint cars fail the physical horizon. These conclusions remain fixed-combination statements.

For fixed inventory and ownership, the load after a visited subset S is $\sum_{i \in S}(b_i - Y_i)$. The classical (S, last) dynamic program therefore covers every elementary order while retaining only $0 \le \text{load}(S) \le Q_k$. A second, unloaded recurrence isolates the tour contribution. Both use exact additions of the supplied dyadic travel coefficients at scale 2^{53} in 128-bit integers, with a common capacity of 16777216 subset/last slots. An exhausted complete recurrence gives a fixed-combination minimum; insufficient capacity gives UNKNOWN. Work is $O(m^2 2^m)$ and storage $O(m2^m)$.

These are inherited classical tools, not new theory. Their precision is bound to the supplied discrete coefficients and acceptance contract, not all symbolic Euclidean calculations. A loaded-tour minimum at an old Y is not transported as a necessary lower for variable new quantities. No new variable-Y tour cut or complete-Q extension ceiling is claimed.

4 Evidence, cost and the stopping boundary

The fixed-owner RX35 review-side point is an inherited control, not a new R159 result. Its car 2 already needs 6412.088151837445 seconds at $T = 6300$; its identical fixed-combination problem is not sent to another heavy oracle. The new joint point changes the physical structure and receives its own finite completion. Fixed-owner failures exclude only their actual inventory/ownership combinations. A fully free-owner fixed-Y oracle is reported in its separately recorded scope; an UNKNOWN excludes nothing.

Original-J fractions, all accepted quantities/owners, per-car balances, nominal-T resource times, active sets, raw full-resource lifts, DP routes and exact costs are retained. Qualification includes actual saved model and Start readback, original RN preservation, objective transport, eligibility and outside bounds, all conditional budgets, duplicate-owner rejection, brute-force prefix-load/empty/over-time fixtures and capacity UNKNOWN. All reviews are separate routines by the same agent; no independent-person review is claimed.

The sole deeper query fixes RX35's new Y and releases every eligible owner, route order and load variable. Assignment is only a Start, confirmed by actual bounds. It returns normally at 8192 nodes in 1329.576 outer seconds (Native 1328.935 seconds, Work 2462.051), with zero solutions and status NODE_LIMIT/UNKNOWN. It excludes neither the fixed Y nor any objective level. The constant-zero feasibility bound is not an original-J lower. No E2 structure branch or E3 whole-level time query is entered.

All scientific outer calls, including qualification, failures, DP, independent readback and the final audit, cost 1438.3466242998838 seconds against 129600 seconds of capacity. Nested Runtime/Work is not added twice. The four failed metadata/readback launches remain with source bytes and closed receipts; no Native optimization failure is hidden. The single compile costs 5.8637514999 seconds separately; writing/render/public I/O has separate receipts. The 36-hour capacity is not a minimum expenditure or a calendar deadline.

No physical incumbent or global original-J lower improves. The two physical reference uppers and inherited inventory lowers remain unchanged. Consequently F is not entered: there is no integrated E+R candidate or empty complete panel, no injected offline assignment, and no formal performance or C++-versus-Python speedup claim. The cheap resource construction is retained as a bounded component, with its tour/prefix conversion obstacles and local owner-necessity evidence. A future online candidate needs a useful qualified own-state route/proof benefit and full self-paid integration. Another unchanged UNKNOWN window or parameter grid is not justified by unused capacity.

R76/R83 already provide activation, quantity closure and equal-net block exchange; R96 already provides full fixed-route quantities and reorder diagnostics. R103 establishes per-car resource hulls. R105–107 retain negative complete evidence for master/oracle/conflict architectures. R157 already uses original-J resource optimization and free-fleet completion; R158 already jointly perturbs inventory and ownership and records a 36-hour quarter-level UNKNOWN. Handling identities, original-J modeling, owner exchange and subset DP are not new contributions here. The present evidence concerns one finite simultaneous inner domain, two same-rule sources, and explicit resource-to-physical obstructions. It does not establish universal novelty, distributional generalization or a complete-method speedup. Stable GI-TBC/B remains the formal entry.